DISTINGUISHING HOMEOMORPHIC 4-MANIFOLDS BY SLICING AND THE EXOTIC $S^2 \times S^2$ PROBLEM

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Abstract. Lidman and Piccirillo constructed the first pair of closed 4-manifolds with isomorphic integer cohomology rings distinguished by unconstrained knot slicing: spin rational homology 4-spheres B and W with $H_1 = \mathbb{Z}/2$ in which the figure-eight knot is slice and not slice, respectively. We show that the natural upgrade of their theorem — a *homeomorphic* such pair — is equivalent to the exotic $S^2 \times S^2$ problem: for every admissible Luttinger-surgery variant V' of their key piece (we classify the admissible parameters), $W' = V'/\sigma$ is homeomorphic to B if and only if the symplectic double $Z' = V' \cup_\sigma V'$ is simply connected, in which case Z' is an exotic $S^2 \times S^2$; if moreover the piece V' itself is simply connected, their regluing additionally yields a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. The proof combines the Hambleton–Kreck classification with a rigidity analysis of the surgery parameters. We then localize what stands between the construction and the goal: building a fully explicit model of the relevant surface-bundle group from the trefoil monodromy, we verify by machine that all 72 uncorrected candidate presentations of $\pi_1(V')$ collapse to the trivial group, and that this collapse is over-determined — carried entirely by four monodromy relations broken by the surgery tori — so that the value of $\pi_1(V')$ rests on the based-loop correction words alone. The same tools verify that the group theory of Akhmedov–Park's claimed exotic $S^2 \times S^2$ (arXiv:1005.3346) is correct given the presentation of their Lemma 8, isolating the open content of that sixteen-year-old preprint in the geometric derivation of its meridian and framing words. The computation itself — deriving the actual based-loop data and deciding the groups — is carried out in a companion paper.

1. Introduction

One approach to distinguishing smooth structures, going back to Casson, is to find a knot that is smoothly slice in one 4-manifold and not in another with the same underlying topology. Lidman and Piccirillo [LP25] recently carried out the first successful version of this program at the level of cohomology rings:

Theorem 1.1 ([LP25, Theorem 1])**.** *There are spin rational homology four-spheres B and W with $H_1 = \mathbb{Z}/2$ and isomorphic integer cohomology rings such that the figure-eight knot 4_1 is slice in B but not in W.*

Here B is the Kawauchi manifold [Kaw09] and $W = V/\sigma$, where V is a symplectic homology $S^2 \times D^2$ built from a genus-2 surface bundle R over a once-punctured torus by two Luttinger surgeries, and σ is a free involution of $\partial V = S^3_0(Q)$, Q the square knot. The natural strengthening — replace "isomorphic cohomology rings" by "homeomorphic", producing the first homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing — requires exactly $\pi_1(W') = \mathbb{Z}/2$ for a variant W' of the construction. Our first result says this is not an incremental improvement but a genuine obstruction, and explains, we believe, why [LP25] stops at cohomology rings.

Date: July 15, 2026.

To state it, call a 4-manifold V' an *admissible variant* of V if it is obtained from R by Luttinger surgeries as in Proposition 1.3 below (these are exactly the surgeries on the tori T_α, T_β and parallel copies producing a homology $S^2 \times D^2$; the LP manifold is the case $(m,n) = (0,0)$ with no extra tori). Set $W' = V'/\sigma$ and $Z' = V' \cup_\sigma V'$.

Theorem 1.2 (Reduction)**.** *Let V' be any admissible variant. Then:*

(a) $W' \cong_{\text{homeo}} B$ *if and only if* $\pi_1(Z') = 1$.
(b) *If* $\pi_1(Z') = 1$, *then* Z' *is homeomorphic to* $S^2 \times S^2$ *and not diffeomorphic to it, and the pair* (B, W') *is homeomorphic and non-diffeomorphic, distinguished by the sliceness of* 4_1.
(c) *If* $\pi_1(V') = 1$ *(which implies* $\pi_1(Z') = 1$*), then in addition the regluing of* [LP25, Theorem 2] *applied to* Z' *yields a simply connected* 4*-manifold homeomorphic but not diffeomorphic to* $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

An exotic $S^2 \times S^2$ — or an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ — is a long-standing open problem; the only claim in the literature, by Akhmedov and Park [AP10], has remained an unpublished preprint since 2010 and is cited as such by the current literature [SS23, BSS24], while [LP25] re-proves the strictly weaker "nonstandard cohomology $S^2 \times S^2$" statement (their Theorem 8) and describes the known examples as "presumably not simply connected". Theorem 1.2 runs in both directions: the slicing upgrade cannot be obtained without solving the exotic $S^2 \times S^2$ problem inside a σ-symmetric double, and conversely any solution of the simple-connectivity problem in the LP frame yields three theorems simultaneously — strictly more than the Akhmedov–Park frame, which yields the exotic $S^2 \times S^2$ alone — with no additional work, since surgeries in $\operatorname{int} V$ descend to the quotient automatically.

The proof of Theorem 1.2(a) rests on the topological classification of 4-manifolds with finite cyclic fundamental group: by Freedman and Hambleton–Kreck [Fre82, HK88, HK93] (see [Ham08, Theorem 5.1] for the statement in this form, and [BSS24] for its $b_2 = 0$ case), a closed oriented topological 4-manifold with $\pi_1 = \mathbb{Z}_n$ is determined up to homeomorphism by its intersection form, w_2-type, and Kirby–Siebenmann invariant; in particular the smooth spin rational homology 4-sphere with $\pi_1 = \mathbb{Z}/2$ is unique up to homeomorphism. The input on the smooth side is a rigidity statement for the surgery parameters:

Proposition 1.3 (Rigidity of the surgery parameters)**.** *Any Luttinger surgeries on $T_\alpha, T_\beta \subset R$ producing a homology $S^2 \times D^2$ have coefficient ± 1, and the surgery directions equal α', β' modulo fiber classes. Conversely, for every $m, n \in \mathbb{Z}$ the directions $\beta' e^m, \alpha' c^n$ with coefficients ± 1 (and, optionally, ± 1-surgeries along parallel Lagrangian copies of T_α, T_β with fiber directions) yield $H_1(V') = 0$ and preserve: the spin condition, symplecticity, $\chi = 2$, $H_2(V') = \mathbb{Z}\langle F \rangle$, the descent of σ, and the hypotheses of* [SS23, Theorem 1.4] *for the double. In particular* [LP25, Lemma 7] *holds verbatim for W', and the slicing obstruction for 4_1 in W' persists.*

The second half of the paper concerns what stands between the construction and $\pi_1(V') = 1$, and makes the obstruction quantitative. In Section 2 we build a fully explicit finite presentation of $\pi_1(R)$ with no genus-2 mapping class computations: since Q is the square knot, the closed genus-2 fiber splits along the connect-sum circle and the entire bundle structure is generated by the trefoil monodromy

$$h\colon x \mapsto y^{-1}, \qquad y \mapsto yx$$

on $F_2 = \langle x, y \rangle$, together with the handle swap. All structural identities (that h is the composite of the two standard twists; that it fixes the boundary word $[x, y]$ exactly; that h^6

is the boundary Dehn twist and h^3 the hyperelliptic involution; that Lidman–Piccirillo's commutator factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ is the literal automorphism identity $\tilde{\psi}\tilde{\varphi}\tilde{\psi}^{-1}\tilde{\varphi}^{-1} = h * h^{-1}$) are certified by machine; the scripts are ancillary files.

With the model in hand, Section 5 reports three computations (GAP 4.16, coset enumeration; all runs reproduce from the ancillary scripts in seconds):

(E1) All 72 *uncorrected* candidate presentations of $\pi_1(V')$ — over all sign and ordering conventions, the LP directions, the mixed directions $m, n \in \{-1, 1\}$, and the parallel-tori scheme — collapse to the trivial group, while the controls ($\pi_1(R)$ alone: $H_1 = \mathbb{Z}^2$; either surgery alone: $H_1 = \mathbb{Z}$) do not. If these presentations were correct, LP's own Z would already be an exotic $S^2 \times S^2$; they are therefore over-determined, and the failure is localized in exactly four monodromy relations, those carried by transport annuli that the surgery tori must intersect.

(E2) Dropping the four broken relations outright leaves $H_1 = \mathbb{Z}^2$: the corrections are indispensable already for homology, so no certificate assembled from the clean relations can bypass them. The value of $\pi_1(V')$ — and with it, by Theorem 1.2, the existence of an exotic $S^2 \times S^2$ — is carried entirely by the based-loop correction words, and a naive enumeration proves nothing in either direction.

(E3) From the presentation printed in [AP10, Lemma 8], coset enumeration confirms $\pi_1(M_n^p) = \mathbb{Z}/p$ for $(n, p) \in \{(1, 1), (2, 1), (3, 1), (1, 2), (1, 3)\}$: the group theory of [AP10, Theorem 9] is correct *given* Lemma 8. What remains open in [AP10] is precisely the geometric derivation of the six meridian/framing words of their Lemma 7 — the same kind of based-loop data as in (E2), and not machine-checkable from a figure.

We read (E2) as the formal reason [AP10] has remained unaccepted for sixteen years: at this level of explicitness, nothing short of a genuine based diagram decides the group. Constructively, the model reduces the obstruction to a finite, well-posed computation — derive the correction words for the broken monodromy relations (four over the embedded curves; a minimal based diagram realizes three) and the two based surgery relations from an octagon picture of the fiber — after which the ancillary harness decides $\pi_1(V')$, and with it three open problems, in seconds.

Finally, the geometric dual of T_α requires a small correction to the proof of [LP25, Lemma 5] (Remark 3.1), which is used throughout; and the based-loop computation posed by (E2) — deriving the actual meridian, lasso and Lagrangian-framing words and deciding the groups — is carried out in the companion paper [W26b].

2. The explicit model of $\pi_1(R)$

We recall the construction of [LP25, Section 1]. The 0-surgery $S_0^3(Q)$ on the square knot $Q = 3_1\#\overline{3_1}$ fibers over S^1 with closed genus-2 fiber F and monodromy conjugate to $abd^{-1}e^{-1}$, a positive Dehn twist along each of the chain curves a, b, c, d, e of [LP25, Figure 1] being denoted by the same letter. Let φ be the involution of F exchanging $a \leftrightarrow e$, $b \leftrightarrow d$ and preserving c setwise. Since $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ in the mapping class group, the F-bundle R over the once-punctured torus with monodromy ab along β and φ along α has $\partial R = S_0^3(Q)$.

Convention 2.1. $[u, v] = uvu^{-1}v^{-1}$. Automorphisms act on the left and compose right-to-left: fg means "g first". Mapping classes act on π_1 only up to inner automorphism; every presentation below depends on a choice of automorphism lifts, see Remark 2.6.

2.1. **The trefoil monodromy.** Let $\Sigma_{1,1}$ be a once-punctured torus with basepoint on the boundary, $\pi_1(\Sigma_{1,1}) = F_2 = \langle x, y\rangle$, boundary word $[x, y]$, and let the core curves realizing x, y be $a \simeq x$ and $b \simeq y$, so $a \cdot b = 1$. For a suitable orientation convention the twists act by

$$T_a \colon (x, y) \mapsto (x,\, yx), \qquad T_b \colon (x, y) \mapsto (xy^{-1},\, y).$$

Define $h := T_a \circ T_b$ (T_b first).

Lemma 2.2. *$h(x) = y^{-1}$, $h(y) = yx$, and:*

1. $h([x, y]) = [x, y]$ exactly *(not merely up to conjugacy);*
2. $h^6 = \mathrm{conj}_{[x,y]^{-1}}$*, the full boundary Dehn twist;*
3. h^3 *acts on* H_1 *as* $-\mathrm{id}$*;*
4. *on* $H_1(\Sigma_{1,1}) = \mathbb{Z}^2$*,* h *acts by* $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$*: determinant* 1*, trace* 1*, order* 6*.*

Consequently h is the fibered monodromy of a trefoil: it is a product of single positive twists along dual non-separating curves, fixes the boundary pointwise (reflected in (1)), and is freely periodic of period 6 with h^6 the boundary twist (2), with the hyperelliptic involution at h^3 (3).

Proof. Finite computations in F_2, certified by the ancillary scripts `monodromy_check2.g` and `model_check3.g`, and easily reproduced by hand. For (1):

$$h([x, y]) = [y^{-1}, yx] = y^{-1} \cdot yx \cdot y \cdot x^{-1}y^{-1} = xyx^{-1}y^{-1} = [x, y].$$

□

Remark 2.3 (Chirality)**.** Whether h or h^{-1} ($h^{-1}\colon x \mapsto xy,\ y \mapsto x^{-1}$) is the right-handed trefoil is a convention we will not need: Q is the connected sum of a trefoil and its mirror, and exchanging the chirality assignment amounts to the handle relabeling $(x, y) \leftrightarrow (r, s)$ below, under which every construction and every computation in Section 5 is symmetric.

2.2. **The closed fiber and the bundle group.** The fiber of $Q = 3_1\#\overline{3_1}$ is the boundary connected sum of the two trefoil fibers, and the closed fiber F of $S^3_0(Q)$ splits along the connect-sum circle into two once-punctured tori. Writing $\pi_1(F) = \langle x, y, r, s \mid [x, y][r, s]\rangle$ with (x, y) carried by the left handle and (r, s) by the right, the dictionary with [LP25, Figure 1] is

$$a \sim x, \quad b \sim y, \quad e \sim r, \quad d \sim s, \qquad c \sim xr, \quad z \sim ys,$$

where the last two identify the φ-invariant homology classes $[c] = [a] + [e]$ and $[z] = [b] + [d]$; the specific words xr and ys are representative choices (see Remark 2.6 and Section 5). Define

$$\tilde{\varphi}\colon\ x \leftrightarrow r,\ y \leftrightarrow s \qquad \text{and} \qquad \tilde{\psi} := h * \mathrm{id}\colon\ (x, y, r, s) \mapsto (y^{-1},\, yx,\, r,\, s),$$

lifting φ and ab respectively; $\tilde{\psi}$ is well defined on the closed-fiber group because h fixes the boundary word exactly (Lemma 2.2(1)).

Lemma 2.4. (1) *$\tilde{\psi}$ fixes the relator $[x, y][r, s]$ exactly; $\tilde{\varphi}$ sends it to its $[x, y]$-conjugate. In particular both define automorphisms of $\pi_1(F)$.*
(2) *$\tilde{\varphi}^2 = \mathrm{id}$, and $\tilde{\psi}\,\tilde{\varphi}\,\tilde{\psi}^{-1}\,\tilde{\varphi}^{-1} = h * h^{-1}\colon (x, y, r, s) \mapsto (y^{-1},\, yx,\, rs,\, r^{-1})$.*

Proof. Machine-certified (`model_check3.g`); (2) also follows from block algebra: $\tilde{\varphi}(h * \mathrm{id})^{-1}\tilde{\varphi} = \mathrm{id} * h^{-1}$, so the commutator is $(h * \mathrm{id})(\mathrm{id} * h^{-1}) = h * h^{-1}$. □

Lemma 2.4(2) realizes the mapping-class factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ of [LP25] as a literal identity of automorphisms: $h * h^{-1}$ is the 0-surgery monodromy of the square knot in this model (the left handle twisted by the trefoil, the right by its mirror).

Proposition 2.5 (The presentation)**.**

$$\pi_1(R) = \big\langle x, y, r, s, \alpha, \beta \;\big|\; [x,y][r,s], \\ \alpha g \alpha^{-1} = \tilde{\varphi}(g),\ \beta g \beta^{-1} = \tilde{\psi}(g) \quad (g \in \{x,y,r,s\}) \big\rangle.$$

Proof. R fibers over the once-punctured torus T_0 with aspherical fiber F; the long exact sequence of the fibration reduces to $1 \to \pi_1(F) \to \pi_1(R) \to \pi_1(T_0) \to 1$, and $\pi_1(T_0) = F_2\langle \alpha, \beta\rangle$ is free, so the sequence splits and $\pi_1(R) = \pi_1(F) \rtimes F_2$ with respect to the chosen automorphism lifts $\tilde{\varphi}, \tilde{\psi}$ of the two monodromies. The displayed presentation is the standard presentation of such a semidirect product. □

Remark 2.6 (Lift freedom; where the based-loop indeterminacy lives)**.** A mapping class determines its action on $\pi_1(F)$ only up to inner automorphisms. Replacing $\tilde{\varphi}, \tilde{\psi}$ by other lifts changes the presentation of Proposition 2.5 by the substitution $\alpha \mapsto \alpha w$, $\beta \mapsto \beta w'$ $(w, w' \in \pi_1(F))$ — an isomorphism of $\pi_1(R)$, but one that changes the *words* that geometric objects (meridians, surgery directions) are represented by. All presentations of the surgered manifolds in Section 5 are therefore *candidates*, well defined only once based representatives are fixed; the dependence of $\pi_1(V')$ on that choice is precisely the point of experiment (E2).

Remark 2.7 (Convention robustness)**.** Replacing the monodromy lift $\tilde{\psi}$ by $\tilde{\psi}^{-1}$ (the holonomy-direction convention) is the substitution $\beta \mapsto \beta^{-1}$, which converts the surgery relation $[z,\alpha]^{\varepsilon}\beta = 1$ into $[z,\alpha]^{-\varepsilon}\beta = 1$: an ε-flip, and the computational grid of Section 5 sweeps all ε. Together with Remark 2.3, the collapse of the LP cases below is independent of every orientation and composition convention made here; the mixed-direction and parallel-tori cases (whose convention variants differ additionally by word order and conjugation) are run in the conventions stated.

3. Rigidity of the surgery parameters

Recall from [LP25, Section 1] the two Lagrangian tori: T_β, the sub-bundle with fiber e over β, and T_α, the sub-bundle with fiber c over α (the monodromy ab fixes e pointwise; φ preserves c with orientation). Their meridians are commutators in the complement $C := R \setminus (T_\alpha \cup T_\beta)$: $\mu_{T_\beta} \simeq [z, \alpha'']$ and $\mu_{T_\alpha} \simeq [d, \beta'']$ (see Remark 3.1), where α'', β'' project to α, β. More generally, for the parallel-copy families below, every relevant meridian is such a commutator: the geometric-dual construction applies verbatim to each parallel copy, whose dual sub-bundle torus meets it once and exhibits the meridian as the corresponding commutator.

Remark 3.1 (The geometric dual of T_α)**.** The proof of [LP25, Lemma 5] prints the dual as (b, β''); this cannot close up into a torus, since ab does not preserve b even homologically (the only twist in ab affecting b is T_a, and $a \cdot b = 1$, so $[ab(b)]$ has $[a]$-coefficient ± 1 in every convention). The curve dual to c that ab *does* fix — pointwise, being disjoint from $a \cup b$, exactly as [LP25, p. 3] observe for e — is d; the dual is therefore $T^*_\alpha = (d, \beta'')$, with μ_{T_α} freely homotopic to $[d, \beta'']$. Since φ exchanges $b \leftrightarrow d$, the confusion is a natural one, and no result of [LP25] is affected: their arguments use only that each meridian is a commutator of a fiber curve with a curve projecting to the base. At the π_1 level, however, the correction matters, and it is used throughout this paper and the companion [W26b].

Lemma 3.2. *Let $C = R \setminus \nu(\mathcal{T})$ for any finite disjoint union $\mathcal{T}$ of tori from the two admissible sub-bundle families. Then the inclusion induces an isomorphism*

$$H_1(C) \xrightarrow{\cong} H_1(R) = \mathbb{Z}^2\langle \alpha', \beta' \rangle.$$

In particular every fiber class vanishes in $H_1(C)$.

Proof. First, $H_1(R) \cong \mathbb{Z}^2$: by Proposition 2.5, $H_1(R)$ is the direct sum of $\mathbb{Z}^2\langle \alpha, \beta \rangle$ and the coinvariants of $H_1(F) = \mathbb{Z}^4$ under the subgroup generated by $\tilde{\varphi}_*$ and $\tilde{\psi}_*$; the stacked matrix $(\Phi - I; \Psi - I)$ has Smith normal form $\mathrm{diag}(1, 1, 1, 1)$ (ancillary script `model_check3.g`), so the coinvariants vanish. (Concretely: $\Psi - I$ is invertible on the (x, y)-block, killing x, y; then the swap relations kill r, s.)

For the complement, excision and the Künneth formula for $(\nu(T), \partial\nu(T)) \cong T^2 \times (D^2, \partial D^2)$ give $H_1(R, C) = 0$ and $H_2(R, C) \cong \bigoplus_{T \in \mathcal{T}} \mathbb{Z}$, generated by the normal disks, whose boundaries are the meridians. The homology sequence of the pair,

$$H_2(R, C) \xrightarrow{\partial} H_1(C) \to H_1(R) \to H_1(R, C) = 0,$$

therefore presents $H_1(R)$ as the quotient of $H_1(C)$ by the images of the meridians; each meridian is a commutator in $\pi_1(C)$, hence trivial in $H_1(C)$, so the map $H_1(C) \to H_1(R)$ is an isomorphism. $\square$

Proof of Proposition 1.3. Forcing. A Luttinger ($1/k$-log) surgery on T_β along an embedded direction, i.e. a primitive class $p\beta' + qe$ on T_β ($\gcd(p, q) = 1$), fills the boundary of $\nu(T_\beta)$ so that the class $\mu_{T_\beta} + k(p\beta' + qe)$ dies (orientation conventions are absorbed into the sign of k). By Lemma 3.2, in H_1 this relation reads $kp\,\beta' = 0$, since the meridian is a commutator and e is a fiber class. Together with the corresponding relation $k'p'\,\alpha' = 0$ from T_α,

$$H_1(V') \;=\; \mathbb{Z}^2/\langle kp\,\beta',\, k'p'\,\alpha' \rangle \;=\; \mathbb{Z}/|kp| \oplus \mathbb{Z}/|k'p'|,$$

which vanishes iff $|kp| = |k'p'| = 1$: the coefficients are ± 1 and the directions have base component ± 1, i.e. equal β', α' modulo fiber classes. Purely-fiber directions ($p = 0$) impose no relation and leave $H_1 = \mathbb{Z}^2$.

Realization. Conversely, for coefficients ± 1 and directions $\beta' e^m, \alpha' c^n$ the same computation gives $H_1(V') = 0$; adding ± 1-surgeries along parallel copies with fiber directions adds only trivial H_1-relations and keeps $H_1(V') = 0$. The admissible parallel families are constrained by the intersection pattern of the fiber curves: with $[c] = [a] + [e]$ and $[z] = [b] + [d]$ one has $c \cdot e = 0$, $c \cdot d = z \cdot e = d \cdot e = 1$, $z \cdot d = 0$, so the maximal disjoint families are $\{c\text{-fibered over } \alpha\text{-parallels}\} \cup \{e\text{-fibered over } \beta\text{-parallels}\}$ (the LP family) or the complementary $\{z/\alpha\} \cup \{d/\beta\}$ family, but no mixture.

Preserved structure. Each direction above is a primitive class realized by an embedded curve on the corresponding Lagrangian torus, so the surgeries are Luttinger surgeries and the result carries a symplectic structure [Lut95, ADK03]; χ is unchanged. The argument of [LP25, Lemma 5] now applies verbatim: $H_1(V') = 0$ and $\partial V'$ connected give $H_3(V') = 0$ and $H_2(V')$ free; $\chi(V') = 2$ gives $H_2(V') \cong \mathbb{Z}$, generated by a fiber F disjoint from all surgery tori; $F^2 = 0$; since $H_1(V') = 0$ we get $H_2(V'; \mathbb{Z}/2) \cong \mathbb{Z}/2\langle F \rangle$, and evaluation against it detects $H^2(V'; \mathbb{Z}/2)$, so $\langle w_2, F \rangle = F \cdot F = 0$ forces $w_2(V') = 0$: V' is spin. All surgeries take place in $\operatorname{int} V$, so $\partial V' = \partial V = S^3_0(Q)$ and the free involution σ descends; the proof of [LP25, Lemma 7] applies verbatim and $W' = V'/\sigma$ is a spin rational homology 4-sphere with $H_1(W') = \mathbb{Z}/2$.

Persistence of the obstruction input. The double $Z' = V' \cup_\sigma V'$ is obtained from the genus-2 bundle $R \cup_\sigma R$ over a genus-2 surface by Luttinger surgeries on disjoint Lagrangian tori; the fiber F and the section $\hat{\Gamma}$ coming from a fixed point of φ ([LP25, Lemma 6]) form a hyperbolic pair, and all surgery tori miss a fiber (they lie over curves in the base, which miss a point) and miss the two fixed-point sections (the fiber curves c, e, z, d and their parallel copies can be isotoped off the two fixed points of φ, exactly as in [LP25, Lemma 6]). These are the hypotheses of [SS23, Theorem 1.4] as used in [LP25, proof of Theorem 1, p. 6], which constrains neither the surgery coefficients nor the directions. Hence the slicing obstruction for 4_1 in W' persists; the details are completed in the proof of Theorem 1.2(b) below. □

4. The reduction theorem

Proof of Theorem 1.2. (a). The double cover $Z' \to W'$ gives a short exact sequence $1 \to \pi_1(Z') \to \pi_1(W') \to \mathbb{Z}/2 \to 1$, with $\pi_1(Z') \hookrightarrow \pi_1(W')$ *injective* by covering theory.

($\Leftarrow$) If $\pi_1(Z') = 1$ then $\pi_1(W') \cong \mathbb{Z}/2$. Now W' and B are closed, oriented, *smooth* spin 4-manifolds with $\pi_1 = \mathbb{Z}/2$ (Proposition 1.3 for W'; [Kaw09, LP25] for B) and both are rational homology spheres, so both intersection forms on H_2/tors are trivial, both w_2-types are type (II), and both Kirby–Siebenmann invariants vanish (smoothness; alternatively $\mathrm{KS} \equiv \sigma/8 \pmod 2$ for spin topological 4-manifolds). By the classification of closed oriented topological 4-manifolds with finite cyclic fundamental group — [HK93, Theorem C]: π_1, the intersection form on H_2/Tors, the w_2-type and KS are a complete set of invariants (extending [HK88, Theorem B] from odd order; see also [Ham08, Theorem 5.1] and [Tei92]) — we get $W' \cong_{\text{homeo}} B$. In fact, by the $b_2 = 0$ case as stated in [BSS24] there are exactly two such topological manifolds for each cyclic group, one spin and one non-spin; the spin one is the spun lens space, so $W' \cong_{\text{homeo}} B \cong_{\text{homeo}} L_2$.

($\Rightarrow$) A homeomorphism gives $\pi_1(W') \cong \pi_1(B) = \mathbb{Z}/2$, and the injection above forces $|\pi_1(Z')| = 1$.

(b). Assume $\pi_1(Z') = 1$.

Step 1: Z' is homeomorphic to $S^2 \times S^2$. Z' is closed, simply connected, spin (it double covers the spin manifold W'; alternatively glue the spin structures of the two copies of V'), with $\chi(Z') = 2\chi(V') - \chi(S^3_0(Q)) = 4$, so $b_2(Z') = 2$. The classes of the fiber F and of the closed section $\hat{\Gamma} = \Gamma \cup_\sigma \Gamma$ satisfy $F^2 = 0$ and $F \cdot \hat{\Gamma} = 1$; since the form is even (spin), a change of basis $\hat{\Gamma} \mapsto \hat{\Gamma} - \frac{1}{2}(\hat{\Gamma}^2)F$ exhibits the intersection form as the hyperbolic form H. By Freedman's classification [Fre82] (even forms are realized by a unique topological manifold), $Z' \cong_{\text{homeo}} S^2 \times S^2$.

Step 2: Z' is not diffeomorphic to $S^2 \times S^2$. This is the argument of [LP25, Theorem 8], which is π_1-agnostic: $R \cup_\sigma R$ is a non-trivial genus-2 surface bundle over a genus-2 surface, hence aspherical, hence minimal and neither rational nor ruled, so its Kodaira dimension satisfies $\kappa \neq -\infty$; Luttinger surgery preserves κ [HL12], and Z' is minimal (it is spin), so $\kappa(Z') \neq -\infty$. Since κ is a diffeomorphism invariant [Li06] and $\kappa(S^2 \times S^2) = -\infty$, Z' is not diffeomorphic to $S^2 \times S^2$.

Step 3: the pair (B, W'). By (a), $W' \cong_{\text{homeo}} B$. The knot 4_1 is slice in B by construction [Kaw09, LP25]. It is not slice in W': following [LP25, Section 2], since W' is spin the relative Rokhlin theorem [Klu21, Theorem 2] and $\mathrm{Arf}(4_1) = 1$ rule out a null-homologous slice disk; a homologically essential slice disk makes the 0-trace $X_0(4_1)$ embed in W' with H_2-essential image, producing a square-zero essential torus $T \subset W'$ (cap a Seifert surface with the disk); since $X_0(4_1)$ is simply connected, T lifts to an essential square-zero torus in Z'; and Proposition 1.3 verifies for Z' the hypotheses of [SS23, Theorem 1.4], which forbids such a torus. Sliceness of 4_1 distinguishes the smooth structures, so W' is homeomorphic and not diffeomorphic to B.

(c). Assume $\pi_1(V') = 1$. The double Z' and the reglued manifold $Z'' = V' \cup_{f\circ\sigma} V'$ (with f the boundary self-homeomorphism of [LP25, Figure 3]) are unions of two copies of V' along $S^3_0(Q)$; by van Kampen each fundamental group is a quotient of $\pi_1(V') * \pi_1(V') = 1$, so $\pi_1(Z') = \pi_1(Z'') = 1$ and parts (a)–(b) apply. (This is where the hypothesis $\pi_1(V') = 1$ is genuinely used: the regluing twists the van Kampen data by f_*, and no implication from $\pi_1(Z') = 1$ alone is claimed; see Remark 4.1.) The framing-parity change makes Z'' non-spin with $b_2 = 2$ and signature 0, so its intersection form is $\langle 1\rangle \oplus \langle -1\rangle$ and Freedman gives $Z'' \cong_{\text{homeo}} \mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. The relative invariants $\Psi_{V',\mathfrak{t}_\pm}$ are non-zero exactly as in [LP25, Lemma 9]: Z' is symplectic and its canonical class evaluates ± 2 on a fiber disjoint from the surgery tori — Luttinger surgery does not change this pairing [ADK03, HL12] — so [OS04] applies as there (cf. [Thu76]). By [LP25, Lemma 10] the mixed invariant of Z'' for the line span$\{F\}$ is non-zero; since $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ (indeed $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2\#D$ for any homology sphere D) admits, for every choice of line, a square-zero splitting along $S^2 \times S^1$ forcing all mixed invariants to vanish ([LP25, proof of Theorem 2], using $HF_{red}(S^2 \times S^1) = 0$), Z'' is not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. □

Remark 4.1. The implication $\pi_1(V') = 1 \Rightarrow \pi_1(Z') = 1$ is immediate from van Kampen; we do not know whether the converse holds in this setting, since in the amalgam $\pi_1(Z') = \pi_1(V') *_{\pi_1 S^3_0(Q)} \pi_1(V')$ the edge maps need not be injective. The distinction affects only part (c) — parts (a) and (b) use $\pi_1(Z')$ alone, while the regluing twists the van Kampen data by f_* and is controlled here through $\pi_1(V')$ — and it is immaterial in practice: any certification of simple connectivity in this frame proceeds by computing $\pi_1(V')$, which is what the harness of Section 5 does.

Corollary 4.2. *The homeomorphic upgrade of* [LP25, Theorem 1] *within the LP frame is* gated*: it holds for some admissible variant if and only if some admissible double Z' is an exotic $S^2 \times S^2$. Conversely, exhibiting $\pi_1(V') = 1$ for any admissible variant yields simultaneously an exotic $S^2 \times S^2$, a homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing, and a simply connected exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$.*

5. Computational experiments

All computations are coset enumerations and abelianizations in GAP 4.16 [GAP]; the scripts are provided as ancillary files,[1] and every run below reproduces in seconds on a laptop (the enumeration cap is 4×10^5 cosets; a run that exceeds it is reported as *blowup* — not visibly trivial, never "nontrivial").

5.1. **Candidate presentations.** By Proposition 2.5 and Remark 2.6, a presentation of $\pi_1(V')$ requires, beyond the exact bundle relations, based representatives for the two surgery relations. The *uncorrected candidates* take the naive words suggested by the free homotopy types: filling relations

$$[z, \alpha]^{\varepsilon_1} \cdot \beta\, r^m = 1, \qquad [w, \beta]^{\varepsilon_2} \cdot \alpha\, (xr)^n = 1,$$

with $z \in \{ys, sy\}$, $\varepsilon_i \in \{\pm 1\}$, mixed-direction exponents m, n as in Proposition 1.3, and, in the parallel-tori scheme, the additional fiber-direction relations $[z, \alpha]^{\varepsilon_3} r = 1$, $[w, \beta]^{\varepsilon_4} (xr) = 1$.

[1] pi1_grid.g, pi1_v2b.g, ap_check.g, monodromy_check2.g, model_check3.g; also available at https://github.com/bwuebben/exotic-s2xs2.

The grid of (E1) takes $w = y$, the dual printed in [LP25]; the corrected dual $w = s$ of Remark 3.1 is run separately.

5.2. **(E1) The uncorrected candidates collapse universally.** Over all 8 conventions for the LP surgeries, the 32 mixed-direction cases ($m, n \in \{-1, 1\}$), and the 32 parallel-tori cases, *every one of the* 72 *uncorrected candidate groups coset-enumerates to the trivial group* (scripts `pi1_grid.g`, `pi1_v2b.g`; under a second), while the controls do not (the bundle group alone has $H_1 = \mathbb{Z}^2$, and either surgery alone leaves $H_1 = \mathbb{Z}$). This collapse cannot be valid: were it, the original LP manifold would already satisfy $\pi_1(V) = 1$ and, by Theorem 1.2, Z would be an exotic $S^2 \times S^2$ — which [LP25] does not claim. The candidates are therefore over-determined, and the failure is localizable a priori. A monodromy relation $\beta g \beta^{-1} = \tilde{\psi}(g)$ is carried by a transport annulus $g \times \beta$; in the surgered manifold the relation holds only if the annulus avoids the surgery tori. Intersection numbers in R decide this: an annulus over one base curve meets a sub-bundle torus over the transverse base curve above their intersection point, and meets a torus over the *same* base curve along it; in both cases the count is the intersection number of the fiber curves. Using $[c] = [a] + [e]$, $[z] = [b] + [d]$:

relation carried over α or β	obstruction
$x\ (\sim a),\ r\ (\sim e)$ over α, β	$a \cdot c = a \cdot e = e \cdot c = 0:$ clean
$y\ (\sim b)$ over β	$b \cdot c = 1:$ crosses T_α
y over α	$b \cdot c = 1:$ crosses T_α
$s\ (\sim d)$ over α	$d \cdot c = 1:$ crosses T_α
s over β	$d \cdot e = 1,\ d \cdot c = 1:$ crosses T_β, T_α

Exactly the four relations for $y \sim b$ and $s \sim d$ are broken; in V' they hold only with insertions of conjugates of words carried by the glued-in tori (c-, e-, α'-, β'-words). The four relations for x and r are clean. (The count is that of membranes over the *embedded* curves. Like the correction words themselves, it is diagram data: in the companion's diagram, whose based base loops meet the embedded curves minimally, only three relations acquire corrections, each at a single crossing.)

5.3. **(E2) The corrections decide the group.** Dropping the four broken relations outright leaves a group with $H_1 = \mathbb{Z}^2$ (script `pi1_v2b.g`): the broken relations are indispensable already for $H_1(V') = 0$, so no surjection from a clean-relations group onto the actual $\pi_1(V')$ can certify anything — the clean group is not even a homology-correct approximation. The value of $\pi_1(V')$ — and with it, by Theorem 1.2, the existence of an exotic $S^2 \times S^2$ — is therefore carried entirely by the based-loop correction words, with no robustness in either direction: a naive enumeration that exceeds the coset cap (a *blowup*) proves nothing, and nothing short of a genuine based diagram decides the group. That computation is carried out in [W26b].

5.4. **(E3) Verification of the Akhmedov–Park group theory.** The manifold of [AP10] is built from the same $\mathbb{Z}/2$-germ: their model surface $(\Sigma_2 \times \Sigma_3)/\mathbb{Z}_2$ fibers over Σ_2 with monodromy the same two-fixed-point involution class as φ. Their Theorem 9 asserts $\pi_1(M_n^p) = \mathbb{Z}/p$ from the presentation printed in their Lemma 8. Coset enumeration from that presentation (script `ap_check.g`) confirms

$$\pi_1(M_n^p) = \mathbb{Z}/p \quad \text{for} \quad (n, p) \in \{(1,1), (2,1), (3,1), (1,2), (1,3)\}.$$

Thus the group theory of [AP10, Theorem 9] is correct *given* their Lemma 8, and the entire content of the sixteen-year-old open status of [AP10] is the geometric derivation of the six meridian/Lagrangian-framing triples of their Lemma 7 (their fifth and sixth, involving a path that is not a sub-bundle coordinate, being the visibly delicate ones). One cannot machine-check a figure; the referee's task for [AP10] and the based-loop computation posed by (E2) are the same task.

6. Discussion

6.1. **The companion computation.** The model of Section 2 reduces the obstruction to a finite, fully specified computation: derive, from an explicit based picture of F carrying the curves $a, \dots, e$, z and the basepoint, the actual correction words for the broken monodromy relations and the based surgery data, in the style of [AP10, Lemma 7] or Baldridge–Kirk [BK08]. Because the crossing pattern of Section 5 is fixed by the intersection combinatorics, a single based diagram parameterizes the entire admissible family of Proposition 1.3 at once. This computation is carried out in the companion paper [W26b], from a rotation-symmetric octagon model in which every monodromy lift is genuinely based and every meridian, lasso and Lagrangian-framing word is derived mechanically from finite combinatorial certificates, and the derivation pipeline is calibrated on two known-answer configurations — a double-Luttinger configuration on T^4 (Baldridge–Kirk) and a Seifert drift calibration built to exercise exactly the monodromy machinery that trivial-monodromy tests cannot; the resulting presentations are proved complete and decided by coset enumeration and Knuth–Bendix completion, with an independently implemented second completion engine confirming the certificates. The outcome is that $\pi_1(V') = 1$ for *all nine* cells $m, n \in \{-1, 0, 1\}$ of the mixed-direction family computed there — including the Lidman–Piccirillo surgery itself — so that, by Theorem 1.2, the double Z of [LP25] is an exotic $S^2 \times S^2$, the pair (B, W) is homeomorphic and non-diffeomorphic, distinguished by the sliceness of 4_1, and the regluing yields a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ — and likewise for every admissible variant. Six cells are decided by coset enumeration. The rest — the enumerations exceed 10^8 cosets on the two hardest cells, which a 56-CPU-hour search had shown to have no nontrivial finite quotient of order at most 10^5, consistent with the group being trivial — are decided by Knuth–Bendix completion, whose confluent rewriting systems reduce every generator to the identity, and whose certificates in fact re-derive all nine cells, in every sign convention, in both based diagrams: enumeration blowup proves nothing in either direction, exactly the caution stated in (E2). We refer to [W26b] for the verification apparatus supporting that computation.

6.2. **An ungated rung.** Theorem 1.2 does not gate every strengthening of [LP25]. Their Theorem 3 produces a 4-manifold A homeomorphic to $\mathbb{CP}^2 \# 5\overline{\mathbb{CP}}^2$ in which the non-vanishing of the mixed invariant for the line $\mathrm{span}\{F\}$ survives; a constrained non-slicing statement on A — no genus-one knot with non-trivial Arf invariant bounds a disk in a class d with $d \cdot F \neq 0$ and $d^2 \geq 0$ — appears provable with the tools already in [LP25, SS23, MMP24], while the unconstrained statement is plausibly false (the classes invisible to a line-based invariant, $d \cdot F = 0$ or $d^2 < 0$, are exactly where Norman disks occur in the standard manifold). We leave this to future work.

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